Evaluating the efficiency and water recovery of direct contact membrane distillation (DCMD) systems with dual heat recovery and recirculation of retentate and permeate

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Abstract

A transient model for hollow-fiber direct contact membrane distillation (DCMD) modules is employed to evaluate a desalination plant in six modes of operation, namely batch, semi-batch, volume-control, salinity-control, continuous, and steady state. The main goal is to reveal DCMD desalination configurations that present the best compromise between maximum energy efficiency and minimum use of brackish water from wells, mainly focusing on applications in sites under hydrological stress, like in the semi-arid Brazilian northeastern region. The obtained results indicate that using an additional heat exchanger for heat regeneration, combined with recirculation of the retentate stream, enhances the system efficiency compared to a scenario that uses only the distillate heat recovery. Additionally, the results also showed that if the main interest is in obtaining high recovery ratios (high volume of produced distillate water per volume of brackish water), the recirculating semi-batch and volume-controlling operating modes are the best alternatives to produce distillated water. However, if the interest is to minimize the thermal energy cost, the best option is the batch operating mode.

Keywords: Desalination; DCMD system; heat recovery; retentate recirculation; energy efficiency.

Nomenclature

Latin Symbols (a–m)		**Latin Symbols (n–z)**	
Symbol	**Description**	**Symbol**	**Description**
a	Specific surface area	N	Molar flux
c_p	Specific heat at constant pressure	N_{fbr}	Number of fibers in the vessel
C	Thermal capacity	Nu	Nusselt number
C_s	Salt concentration	p	Pressure
d	Diameter	Pr	Prandtl number
D	Diffusivity / diffusion coefficient	q_c	Heat transferred to the cold sink
f	Friction factor	q_h	Heat transferred from the hot source
h	Convective heat transfer coefficient	q_{HX}	Heat transferred in the heat exchanger
h_{fg}	Enthalpy / heat of vaporization	Q	Rate of heat transfer
j_{wl}	Distillate flow rate	r	Radius
k	Thermal conductivity	R	Universal gas constant
L	Hollow-fiber module length	Re	Reynolds number
m	Mass	T	Temperature
$\dot{m}$	Mass flow rate	V	Volume
M	Molar mass	$\dot{V}$	Volumetric flow rate
		u	Stream velocity
		z	Position along the module
Greek Symbols		**Subscripts / Superscripts**	
Symbol	**Description**	**Symbol**	**Description**
α	Membrane effectiveness (salt)	a	Relative to air
γ	Relative to lumen or shell phase	amb	Relative to environment
δ	Membrane thickness	col	Relative to collector tank
ε	Porosity	$dist$	Relative to distillate
ε_{HX}	Heat exchanger efficacy	e	Relative to inlet
η	Efficiency	ext	Relative to external
λ	Rejection fraction, retentate stream	l	Relative to lumen
μ	Dynamic viscosity	$m/memb$	Relative to membrane
ρ	Density	mix	Relative to mixture
τ	Tortuosity	o	Relative to outlet
ϕ	Salinity	s	Relative to shell
		vap	Relative to vapor
		w	Relative to water
		$well$	Relative to well
		$\bar{(\cdot)}$	Indicates averaged value

Nomenclature (continued)

Acronyms			
Symbol	**Description**	**Symbol**	**Description**
AGMD	Air Gap Membrane Distillation	PGMD	Permeate Gap Membrane Distillation
CGMD	Conductive Gap Membrane Distillation	RO	Reverse Osmosis
DCMD	Direct Contact Membrane Distillation	RR	Recovery Ratio
DLR	Distillate Limiting Ratio	SEC	Specific Energy Consumption
GOR	Gained Output Ratio	SGMD	Sweeping Gas Membrane Distillation
FLR	Feed Limiting Ratio	V-AGMD	Vacuum Air Gap Membrane Distillation
HICO	Hot In, Cold Out	VMD	Vacuum Membrane Distillation
HOCI	Hot Out, Cold In		
HX	Heat Exchanger		
MD	Membrane Distillation		

1 Introduction

Freshwater scarcity, mainly due to population growth, has attracted the attention of researchers and industry for the development and improvement of techniques for the treatment, purification, and desalination of salt water. Among the main desalination techniques, those involving thermal separation methods, such as membrane distillation, require more energy due to the need to vaporize the water since the water vaporization enthalpy is about 2,400 kJ/kg [Sharqawy et al., 2010], while Gibbs free energy for seawater, which governs the energy use of the reverse osmosis method, is 2.7 kJ/kg [Elimelech and Phillip, 2011]. Therefore, one of the challenges that must be overcome for the application of thermal methods such as membrane distillation (MD) in water desalination, which aims to be competitive against the conventionally employed reverse osmosis (RO) process in small and medium-sized systems (< 100 m^3/day of water produced), is to minimize the demand for thermal energy, improving thermal efficiency.

The membrane distillation process enables the use of residual heat from other thermal processes or the heat available from renewable sources, such as solar collectors. However, heat losses by conduction through the membrane further increase the thermal consumption, which, when added to high-water consumption, limits the large-scale use of these systems [Khayet and Matsuura, 2011]. Nevertheless, according to Guan et al. [2015], when compared to conventional desalination processes, such as multi-stage flash distillation, multi-effect distillation, and reverse osmosis, the MD process has several benefits: low salinity sensitivity and high salt rejection (100% content, theoretically), low membrane vulnerability to fouling, greater performance under low severity operating conditions, and the viability of using low-grade waste heat from other thermal processes or renewable sources, such as solar collectors, as previously mentioned [Curcio and Drioli, 2005, Charcosset, 2009, Drioli et al., 2015].

Concerning its technology, according to the method of obtaining the vapor pressure gradient through the membrane, the MD is classified in the following configurations [Zhang et al., 2015, Swaminathan et al., 2018]: DCMD (Direct Contact Membrane Distillation), AGMD (Air Gap Membrane Distillation), SGMD (Sweeping Gas Membrane Distillation), VMD (Vacuum Membrane Distillation), PGMD (Permeate Gap Membrane Distillation), CGMD (Conductive Gap Membrane Distillation), V-AGMD (Vacuum-Enhanced Air Gap Membrane Distillation), among other variants. Each configuration has its advantages and intrinsic disadvantages, so the choice of one of them is still a research theme, which is usually directed through metrics such as GOR (Gained Output Ratio) and SEC (Specific Energy Consumption), which are parameters for energy efficiency analysis. The DCMD is the most studied and straightforward configuration among them, widely used for desalination studies and concentration of aqueous solutions.

In general, the increase in the thermal efficiency of MD systems is achieved by improving the performance of membranes (by choosing membranes with low thermal conductivity, adequate porosity, pore size and pore size distribution, and thickness), as well as by reducing the amount of external heat to be supplied to the process [Gostoli and Sarti, 1998, Khayet et al., 2006, Al-Obaidani et al., 2008, Yao et al., 2013, Li et al., 2014, Swaminathan et al., 2018].

The heat supplied to the process can then be significantly reduced by the reuse of the heat available in the output currents of the desalination module through external heat exchangers, extending the applicability of the MD process and improving energy efficiency [Gilron et al., 2007, Lee et al., 2011, Hausmann et al., 2012, Lu and Chen, 2012, Zhang et al., 2015, Gonzalez-Bravo et al., 2015, Kim et al., 2015, Maheswari et al., 2015, Lin et al., 2014, ?, Christie et al., 2020, Lisboa et al., 2021, Sampaio,

2022]. Obviously, operating conditions such as temperature, the feed and distillate flow rates, salinity of the feed stream, and its recirculation, among others, significantly affect the MD process and can improve or reduce process efficiency [Khayet et al., 2000, Wang et al., 2009, Zhang et al., 2010, Fan and Peng, 2012, Alkhudhiri et al., 2012, Guan et al., 2014].

Additionally, since MD presents a low recovery ratio per pass (ratio between the production rate of pure water and the feed flow), using some form of recirculation of the brine/retentate stream is mandatory to increase efficiency. In this way, by considering a CGMD module, Swaminathan and Lienhard [2018] describe four operation strategies that enable high global recovery using a single-pass low recovery process (batch recirculation, semi-batch recirculation, continuous recirculation, and continuous multi-stage recirculation). The first two operations are classified as transient/discontinuous processes, while the last two are steady state ones. According to these authors, who studied the salinity variation from a lower limit of 70 g/kg to an upper limit of 260 g/kg, the batch operation is the best option since GOR is about twice for the same distillate flow when compared to the continuous recirculation operation.

Sampaio [2022] modeled a seawater desalination DCMD plant with heat recovery of the recirculating hot distillate stream by using a DCMD module composed by a bundle of hollow-fiber tubes. He combined analytical solutions for the mass transfer problem across the membranes with the finite volume method for solution the steady-state heat transfer problem. Still, based on Christie et al. [2020] and Lin et al. [2014], and to extend their works, Sampaio [2022] developed an approximation for the optimal value of the ratio between permeate and feed flow rates, and for the maximum mass recovery rate that, differently from the expressions developed by Christie et al. [2020] and Lin et al. [2014], take into consideration the membrane thermal efficiency.

Thus, based on the works of Swaminathan et al. [2018], Christie et al. [2020], Lisboa et al. [2021], and Sampaio [2022] and extending their analyses, the present work aims at the development of physical-mathematical modeling, in the unsteady-state regime, and subsequent numerical simulation of a hollow-fiber DCMD system with double recirculation and heat recovery. The main goal is firstly to evaluate the potential of heat recovery from the retentate stream to increase the energy efficiency of the system to produce distilled water. Furthermore, an application of the model to study the desalination process in regions with high seasonal hydrological stress, such as the semi-arid region of the Brazilian northeastern. In addition to the classical batch, semi-batch, continuous, and steady-state modes of operation, two intermediary modes, namely, mixing tank volume controlling and mixing tank salt concentration controlling, are also introduced. The aim is to verify their effect on the behavior of the system efficiency and on the water recovery ratio (for minimizing the well brackish water consumption). These controlling schemes can also be operated in conjunction with total or partial recirculation, i.e., when the brine is rejected from the system with fraction rejections equal or different to the unity, respectively.

Table 1 presents, in a condensed way, a brief bibliographic review of the main experimental and simulation works on the thermal efficiency of MD systems, highlighting the main contributions of each of the works, especially, but not limited to, those works that adopt heat recovery and recirculation of flow streams using hollow fiber membrane configuration.

Table 1: Literature review on thermal efficiency in MD systems by heat recovery.

Reference	Heat Recovery			Observations
	Distillate	Distillate	Retentate	
Fane *et al.* (1987)	X	X	X	Experimental/Simulation – Hollow fiber – DCMD
Schneider *et al.* (1988)	X	X	X	Experimental – Hollow Fiber – DCMD
Kurokawa and Sawa (1996)	X	X	X	Simulation – Flat fiber – DCMD
Li and Sirkar (2004)	—	—	—	Experimental – Hollow Fiber – DCMD – Crossflow
Gilron *et al.* (2007)	X	X	X	Simulation – Hollow fiber – DCMD Cascade
Lee *et al.* (2011)	X	X	X	Experimental/Simulation – Hollow Fiber – DCMD Cascade
Hausmann *et al.* (2012)	—	X	X	Experimental/Simulation – Flat Fiber
He *et al.* (2013)	X	X	X	Simulation – Hollow fiber – DCMD Cascade
Chung *et al.* (2014)	X	X	X	Experimental/Simulation – Hollow fiber – DCMD cascade
Duong *et al.* (2015)	—	X	X	Experimental/Simulation – Flat Fiber – DCMD
Lin *et al.* (2014)	X	X	—	Simulation – DCMD – FLR/DLR
Guan *et al.* (2015)	X	X	X	Simulation – Hollow fiber – DCMD
Zhang *et al.* (2015)	X	X	X	Review on heat recovery in MD systems
Lokare *et al.* (2018)	—	X	X	Simulation – DCMD – Brine recycling
Christie *et al.* (2020)	X	X	X	Simulation – DCMD – FLR/DLR
Lisboa *et al.* (2021)	X	—	—	Simulation – Hollow fiber – DCMD
Sampaio (2022)	X	X	—	Simulation – Hollow fiber – DCMD

2 System Description

The MD desalination plant modelled in the present work is characterized by a water desalination system that operates in the transient regime, containing a DCMD module (a vessel with a bundle of hollow fiber membranes), two tanks (a mixing and collector), and running with dual recirculation and dual heat recovery, where the heat available at the output of the module from the feed and distillate streams are recovered through the heat exchangers, HX-1 and HX-2, respectively, as shown in Figure 1. The heat source and sink are constant-temperature heat reservoirs, so the feed and distillate streams reach the specified values of the source/sink temperatures by supplying/extracting exactly the corresponding amount of heat required.

Figure 1 – Design of the DCMD system with double heat recovery and partial/total recirculation of the retentate current

Figure 1: Design of the DCMD system with double heat recovery and partial/total recirculation of the retentate current.

At the initial time, the feed circuit is completely filled with brackish water at the well salinity (c_s^{well}) and ambient temperature (T_{amb}). The distillate circuit is filled with distilled water at the same ambient temperature. The feed stream leaves the mixing tank at state 3 (c_{s3}, $V_3 = V_{feed}$, T_3) and enters HX-1, which heats it to state 4 (c_{s4}, V_4, T_4) by receiving energy from the distillate leaving the DCMD module at state 11 (c_{s11}, V_{11}, T_{11}). The mandatory use of this heat exchanger (HX-1) greatly increases the efficiency of the MD system by reducing the amount of heat (q_h) that must be supplied by the heater to raise the feed stream temperature to the working temperature at state 5 (c_{s5}, V_5,

$T_5 = T_{feed}$).

The concentrated module effluent (the retentate) at state 6 (c_{s6}, V_6, T_6) can be removed from the system (as a reject stream) or it can be recycled (recirculated). In the present work, the computational code was built taking into consideration both possibilities, simultaneously or not, depending on a flow rejection fraction (λ) at state 9 (c_{s9}, V_9, T_9), which directly and dynamically impacts the concentration of the solution in the tank and its volume, V_{mix}. When $\lambda = 1$, the brine is completely rejected at state 9 (the MD system must be continuously fed from the well on a permanent basis). Otherwise, when $\lambda < 1$, the retentate is partially recycled/rejected.

When necessary, depending on salinity and volume of the mixing tank (or the operating regime), brackish water can be pumped from the well at state 1 (V_{well}, c_s^{well}, T_{well}), passing through HX-2 as a cold stream to gain energy carried by the effluent retentate from the DCMD module at state 6, which will be cooled down to state 7 (c_{s7}, Q_7, T_7). The stream at state 7 can be directed, either totally ($\lambda = 0$) or partially ($0 < \lambda < 1$), to the mixing tank through state 8 (c_{s8}, V_8, T_8), to be mixed with the heated brackish water stream coming from state 2.

To perform the present study, the operational ranges of the PLICA-7 DCMD module produced by Aquastill for experiments in a relevant environment are adopted.

3 Mathematical Model

The implemented mathematical model incorporates heat and mass transfer in all system components: mixing tank, distillate tank, DCMD module, heat source and sink, and heat exchangers 1 and 2. For the DCMD module, the model proposed by Lisboa et al. [2021] was extended to the unsteady-state regime. The authors treated the hollow-fiber DCMD module as an equivalent porous medium composed of the following constituents: the porous shell medium (where the feed stream flows), the porous lumen medium (where the distillate stream flows), and the membrane (where the distillation process occurs).

An illustrative figure of the porous media model employed to analyze the heat and mass transfer in the vessel is shown in Figure 2.

Figure 2 – Illustration of the porous media models.

Figure 2: Illustration of the porous media models. Cross sections of the: (a) Porous medium, lumen side; (b) Module and hollow-fiber membrane bundle; (c) Porous medium, shell side. Figure 1 from Lisboa et al. [2021]. Reprinted with permission from Elsevier. Copyright 2021.

The hollow fiber membranes are considered to be uniformly distributed along a vessel (module) cross section, with their axis parallel to the vessel axis. The module porosity is defined as the ratio of the phase volume to the total vessel volume:

$$\varepsilon_l = \frac{N_{fbr}\,\pi r_l^2}{\pi r_v^2} \tag{1}$$

$$\varepsilon_s = \frac{\pi r_v^2 - N_{fbr}\,\pi r_s^2}{\pi r_v^2} \tag{2}$$

Here, ε_γ is the porosity of phase γ (with $\gamma = l$ for the lumen phase and $\gamma = s$ for the shell phase), N_{fbr} is the number of fibers in the vessel, r_γ is the radius of the γ phase, and r_v is the radius of the vessel. Given that the membrane has a thickness δ, the outer radius of the membrane will be $r_s = r_l + \delta$.

Figure 3 – Representation for the single-fiber model.

Figure 3: Representation for the single-fiber model of the hollow-fiber membrane module.

The specific surface area, a_γ, used in heat and mass transfer calculations, is the ratio of the surface area available for transport phenomena in each phase to the vessel volume:

$$a_\gamma = \frac{N_{fbr}\, 2\pi r_\gamma}{\pi r_v^2} \tag{3}$$

Considering that the heat exchangers, the distillate and mixing tanks, and the DCMD module are adiabatic, and that the salinity and temperature fields inside them are uniform (except inside the DCMD module), the physical situation is mathematically modelled as follows.

Mixture and Distillate Tanks

$$\frac{dV_{mix}}{dt} = \dot{V}_2 + \dot{V}_8 - \dot{V}_3 \tag{4}$$

$$\frac{d(C_{s,mix}\, V_{mix})}{dt} = \dot{V}_2\, C_{s2} + \dot{V}_8\, C_{s8} - \dot{V}_3\, C_{s3} \tag{5}$$

$$\frac{d(T_{mix}\, V_{mix})}{dt} = \dot{V}_2\, T_2 + \dot{V}_8\, T_8 - \dot{V}_3\, T_3 \tag{6}$$

$$\frac{d(T_{col}\, V_{col})}{dt} = \dot{V}_{12}\, T_{12} + \dot{V}_{jwl}\, T_{l,o} - \dot{V}_{11}\, T_{11} \tag{7}$$

DCMD Module

$$\rho_s\, c_{p,s}\, \varepsilon_s\, \frac{\partial T_s}{\partial t} = k_s\, \varepsilon_s\, \frac{\partial^2 T_s}{\partial z^2} - h_s\, a_s\, (T_s - T_{m,s}) - j_{wl}\, a_l\, h_{fg} \tag{8}$$

$$\rho_l\, c_{p,l}\, \varepsilon_l\, \frac{\partial T_l}{\partial t} = k_l\, \varepsilon_l\, \frac{\partial^2 T_l}{\partial z^2} + h_l\, a_l\, (T_{m,l} - T_l) \tag{9}$$

with initial conditions:

$$T_s(z,0) = T_l(z,0) = T_{amb} \tag{10}$$

and boundary conditions:

$$T_s(0,t) = T_5\,, \quad T_l(L,t) = T_{col}(t) \tag{11}$$

$$\left.\frac{\partial T_s}{\partial z}\right|_{z=L} = 0\,, \quad \left.\frac{\partial T_l}{\partial z}\right|_{z=0} = 0 \tag{12}$$

In these expressions, ρ_γ is the specific mass of phase γ, $c_{p,\gamma}$ is the specific heat at constant pressure, k_γ is the thermal conductivity, T_γ is the bulk temperature, $T_{m,\gamma}$ is the interface temperature between

the hollow fiber membrane surface and the liquid phase γ, and h_γ is the convection heat transfer coefficient. The convective heat transfer coefficients for the hot and cold streams are evaluated through the following empirical correlations:

$$Nu_\gamma = \frac{(f_\gamma/8)\, Re_\gamma\, Pr_\gamma}{1.07 + 12.7\sqrt{f_\gamma/8}\,(Pr_\gamma^{2/3} - 1)} \tag{13}$$

where Re_γ is the Reynolds number, Pr_γ is the Prandtl number, and f_γ is the friction factor:

$$Re_\gamma = \frac{\rho_\gamma\, u_\gamma\, d_{h,\gamma}}{\mu_\gamma}\,, \quad d_{h,l} = 2r_l\,, \quad d_{h,s} = 2(r_c - r_s) \tag{14}$$

$$f_l = (0.790 \ln Re_l - 1.64)^{-2}\,, \quad f_s = (0.790 \ln Re_s - 1.64)^{-2} \tag{15}$$

The energy and mass balances for the remaining control volumes in the feed and distillate streams provide the following relationships.

For the temperatures:

$$T_1 = T_{well}\,, \quad T_3 = T_{mix} \tag{16}$$

$$T_6 = T_s(L,t)\,, \quad T_{11} = T_l(0,t) \tag{17}$$

$$T_5 = T_{feed}\,, \quad T_{12} = T_{col} \tag{18}$$

$$T_9 = T_7 \tag{19}$$

$$T_{col,out} = T_{col}\,, \quad T_{mix,out} = T_{mix} \tag{20}$$

For the flow rates:

$$\dot{V}_1 = \dot{V}_{well}\,, \quad \dot{V}_3 = \dot{V}_s \tag{21}$$

$$\dot{V}_5 = \dot{V}_3 = \dot{V}_6 = \dot{V}_s \tag{22}$$

$$\dot{V}_9 = \lambda\, \dot{V}_7\,, \quad \dot{V}_8 = (1-\lambda)\, \dot{V}_7 \tag{23}$$

$$\dot{V}_{11} = \dot{V}_{12} = \dot{V}_l \tag{24}$$

For the salinities:

$$C_{s1} = C_{s,well}\,, \quad C_{s3} = C_{s,mix} \tag{25}$$

$$C_{s6} = C_{s7} = C_{s8} = C_{s9} \tag{26}$$

$$C_{s,col} = 0\,, \quad C_{s,dist} = 0 \tag{27}$$

The temperatures T_4, T_{12} and T_2, T_7 are obtained from the definitions of the effectiveness of heat exchangers HX-1 and HX-2, respectively:

$$\varepsilon_{HX1} = \frac{T_4 - T_3}{T_{11} - T_3} \tag{28}$$

$$\varepsilon_{HX2} = \frac{T_6 - T_7}{T_6 - T_1} \tag{29}$$

$$T_2 = T_1\,, \quad T_{12} = T_{col} \tag{30}$$

In the salinity equations, α_{memb} is the effectiveness of the membrane in retaining salt, taken as 100% in the present study ($\alpha_{memb} = 1$). The properties of the brackish well water are determined in each point of the system as a function of temperature and salinity [Sparrow, 2003].

The temperatures of the membrane interfaces along the z-direction, $T_{m,l}(z,t)$ and $T_{m,s}(z,t)$, required in Eqs. (8)–(9), are evaluated iteratively through the following procedure:

$$Q_m = h_s\, a_s\, (T_s - T_{m,s}) = Q_{cond} + Q_{lat} \tag{31}$$

$$Q_{cond} = \frac{k_m}{\delta}\, a_l\, (T_{m,s} - T_{m,l}) \tag{32}$$

$$Q_{lat} = j_{wl}\, a_l\, h_{fg} \tag{33}$$

$$Q_{conv,l} = h_l\, a_l\, (T_{m,l} - T_l) \tag{34}$$

$$N_{wl} = \frac{j_{wl}}{M_w} \tag{35}$$

The density of the mixture at each position or control volume of the circuit, ρ_i, is related to the salinity, ϕ_i, through the density of pure water and salt at $T_{amb} = 25°\text{C}$ ($\rho_{salt} = 2165$ kg/m^3 and $\rho_{H_2O} = 997.4$ kg/m^3):

$$\rho_i = \rho_{H_2O} + \phi_i\, (\rho_{salt} - \rho_{H_2O}) \tag{36}$$

Other fluid properties, such as specific heat at constant pressure, thermal conductivity, viscosity, and enthalpy of vaporization, are also evaluated as a function of salinity and temperature, according to the correlations proposed by Sparrow [2003].

3.1 Modeling Mass Transfer in the Membrane

To partially close the system of equations, the mass transfer inside the DCMD module must be modeled in order to determine the water vapor mass flux through the membrane, j_{wl}. For this, as performed by Lisboa et al. [2021], the Dusty Gas model [Lawson and Lloyd, 1997], one of the most widely used models for predicting flow of distilled water through membranes, was employed:

$$N_w + \frac{p_a}{p_w}\, \frac{N_w\, D_{wk}}{D_{wa0}} = -\frac{D_{wk}}{R\, T_m}\, \frac{dp_w}{dz} \tag{37}$$

Air is normally considered at rest in the membrane pores, so that $N_a \approx 0$. Applying Dalton's law, one obtains the following expressions for the molar, N_{wl}, and mass, j_{wl}, fluxes of distilled water through the hollow fiber membranes (based on r_l):

$$N_{wl} = \frac{D_{eff}}{R\, T_m\, \delta}\, (p_s - p_l) \tag{38}$$

$$j_{wl} = M_w\, N_{wl} \tag{39}$$

The Knudsen diffusivity of water vapor, D_{wk}, the molecular diffusivity of water in air, D_{wa0}, and

the effective molecular diffusivity, D_{eff}, are evaluated using the classical relationships:

$$D_{wk} = \frac{d_p}{3}\sqrt{\frac{8RT_m}{\pi M_w}} \tag{40}$$

$$D_{wa0} = 1.895 \times 10^{-5}\, T_m^{2.072}\, P^{-1} \tag{41}$$

$$D_{eff} = \frac{\varepsilon_m}{\tau_m}\frac{D_{wk}\, D_{wa0}}{D_{wk} + D_{wa0}} \tag{42}$$

The total pressure inside the membrane pores, P, is usually the atmospheric pressure. The partial vapor pressures of the salt streams at the membrane interfaces, p_s and p_l, are evaluated from the saturated vapor pressures, $p_{v\gamma}$, obtained by using the Antoine equation, corrected for the respective salt mole fractions ($x_{salt,\gamma}$) and activity coefficients (α_γ):

$$p_s = \alpha_s\,(1 - x_{salt,s})\, p_{v,s}(T_{m,s}) \tag{43}$$

$$p_l = \alpha_l\,(1 - x_{salt,l})\, p_{v,l}(T_{m,l}) \tag{44}$$

The most common models for the membrane thermal conductivity are based on geometrical distributions of the solid and gaseous phases. García-Payo and Izquierdo-Gil [2004] recommend that, for porous media with porosity greater than 0.6, the Maxwell I thermal conductivity model be used. These models are mathematically described in terms of the solid matrix, k_s, and fluid/gaseous, k_f, conductivities, as:

$$k_m = \varepsilon_m\, k_f + (1 - \varepsilon_m)\, k_s \quad \text{(parallel model)} \tag{45}$$

$$k_m = \left[\frac{\varepsilon_m}{k_f} + \frac{1 - \varepsilon_m}{k_s}\right]^{-1} \quad \text{(series model)} \tag{46}$$

$$k_m = k_f\,\frac{k_s + 2k_f - 2\varepsilon_m(k_f - k_s)}{k_s + 2k_f + \varepsilon_m(k_f - k_s)}, \quad k_f = \varepsilon_m\, k_{gas} + (1 - \varepsilon_m)\, k_{polymer} \quad \text{(Maxwell I model)} \tag{47}$$

A significant number of tortuosity models can be found in the literature [Ghanbarian et al., 2013]. Kim et al. [2021] analyzed 11 models of tortuosity. The most common ones are:

Geometric (Euclidian) Models:

$$\tau_m = \frac{(2 - \varepsilon_m)^2}{\varepsilon_m} \quad \text{Mackies and Meares (1955)} \tag{48}$$

$$\tau_m = 1 + \frac{1 - \varepsilon_m}{\varepsilon_m}\ln\frac{1}{1 - \varepsilon_m} \quad \text{Beeckman (1990)} \tag{49}$$

Fractal Models:

$$\tau_m = \varepsilon_m^{-0.5} \quad \text{Elias-Kohav } \textit{et al.} \text{ (1991)} \tag{50}$$

$$\tau_m = \varepsilon_m^{-0.5} \quad \text{Bruggeman (1935)} \tag{51}$$

According to Curcino et al. [2022], an adequate selection must be supported on the knowledge of the morphological and structural properties of the membrane used in the experimental desalination module.

3.2 Efficiency Metrics

To evaluate the desalination system and verify the improvement in energy efficiency, the thermal efficiency η_{th}, the gain-output ratio GOR, the specific thermal energy consumption SEC_{th}, and the recovery ratio RR are employed and analyzed in different operational scenarios. Those quantities are defined as:

$$\eta_{th} = 1 - \frac{q_c}{q_h}\,, \quad GOR = \frac{h_{fg}\,\dot{m}_{dist}}{q_h}\,, \quad SEC_{th} = \frac{q_h}{\dot{V}_{dist}}\,, \quad RR = \frac{V_{dist}}{V_{well}} \tag{52}$$

where:

$$q_h = \dot{m}_5\, c_{p5}\,(T_5 - T_4) \tag{53}$$

$$q_c = \dot{m}_{12}\, c_{p12}\,(T_{12} - T_{col}) \tag{54}$$

$$q_{HX1} = \dot{m}_{11}\, c_{p11}\,(T_{11} - T_{12}) \tag{55}$$

$$V_{dist} = \int_0^{t_{end}} \dot{V}_{jwl}\, dt\,, \quad V_{well} = \int_0^{t_{end}} \dot{V}_{well}\, dt \tag{56}$$

The global GOR and SEC_{th} parameters are taken as the average values up to the simulation end time.

A computational code to solve the coupled system of ordinary and partial differential equations was developed in the Fortran 95 language. The MMOLCH subroutine [IMSL Fortran Math Library, 2022], for solving hyperbolic partial differential equation systems through the method of lines, was employed, with a tolerance error set as 10^{-5}. The non-linear equations for the interface temperatures were solved, within an error target prescribed at 10^{-11}, with the routine ZBREN [IMSL Fortran Math Library, 2022].

4 The System Operating Modes

To control the operation of the dynamic system and optimize the desalination system water production and brackish water consumption, a general algorithm based on volume and salinity of the mixture in the mixing reservoir is applied at each instant of time. The developed computational code is very flexible, allowing simulations of different desalination system operation scenarios: steady-state regime (no recirculation), batch recirculation, semi-batch recirculation, continuous recirculation (dynamic rejection fraction/recirculation), and partial recirculation operation mode with volume or concentration control.

For the unsteady-state regimes, the system can start the operation with no brine rejection (total recirculation), $\lambda = \lambda_0 = 0$, or with a pre-defined rejection (partial recirculation), $\lambda_0 \neq 0$. For these regimes, the desalination system operates in such a way that the mixing tank, initially with a volume of brackish water of $V_{mix_0} = 100$ L, has its mixture consumed by the DCMD module, as distillate is produced over time.

Additionally, to analyze the influence of the external heat exchangers on the system performance it is sufficient to vary their effectiveness (ε_{HX1} and ε_{HX2}). By choosing $\varepsilon_{HX1} = 0$ and/or $\varepsilon_{HX2} = 0$, the heat exchangers are not included in the analysis.

4.1 Batch and Semi-Batch Operations

In batch and semi-batch operation modes, over each process cycle, the distillate is continuously removed from the system, and the remaining solution that leaves the MD module, with higher salinity, is directed to the feed tank ($\lambda = 0$). In the batch operation process, there is no water pumping from the well ($\dot{V}_{well} = 0$) between recycling, so the volume of the solution in the tank decreases, and the salt concentration increases over time. In the semi-batch operation cycle, the volume of the mixing tank is kept constant, by continuously adding fresh feed water from the well to the system, at the same proportion the distillate is removed ($\dot{V}_{well} = \dot{V}_{jwl}$).

In both operation processes, the permeate production rate, as well as the heat transfer rate, will reduce as the feed stream of the MD module becomes increasingly saturated with salts. After the salinity of the solution in the mixing tank reaches a limiting value, the high-salinity brine is flushed-out (λ is set to 1) until a minimum volume allowed is reached. Then, the power supply refills the tank from the well ($\dot{V}_{well} = \dot{V}_s$) until the minimum concentration is reached, and the cycle restarts.

4.2 Continuous Operation (Steady-State Operation with Partial Brine Recirculation)

In the continuous operation regime, a fixed operational brine salinity is specified at the MD module exit. Therefore, after the initial transients have elapsed and the solution that leaves the module is at the required salinity (reaching the steady-state regime), it must be continuously bled-out of the system at a specific fraction rejection (λ) that depends on the MD module performance. The make-up water from the well ($\dot{V}_{well}$) is simultaneously added to the tank at a proportional flow rate that keeps constant the salinity of the stream at the MD module exit, which is related to that fraction rejection.

The rejection fraction and the volumetric flow rate from the well is obtained according to the following salt and mixture mass balances at the mixing tank:

$$\lambda = 1 - \frac{\dot{V}_{jwl}\, C_{s,well}}{(\dot{V}_s - \dot{V}_{jwl})\,(C_{s6} - C_{s,well})} \tag{57}$$

$$\dot{V}_{well} = \dot{V}_{jwl} + \lambda\,(\dot{V}_s - \dot{V}_{jwl}) \tag{58}$$

4.3 Steady-State Operation with No Brine Recirculation

To simulate a steady state condition with no recirculation, it is sufficient to set $\lambda = \lambda_0 = \lambda_1 = 1$ and $\dot{V}_{well} = \dot{V}_s$ all the time during the simulations. In this situation, all retentate leaving the DCMD module is rejected and the system is continuously fed by the well at constant feed flow rate and salinity.

4.4 Volume or Concentration Controlling

In the mixing tank volume controlling operation, a minimum and a maximum allowed mixing tank volume is set as the operation limits. The salinity of the water in the mixing tank ($c_{s,mix}(t)$) is continuously increasing with the distillate production, and its volume ($V_{mix}(t)$) decreases. When the minimum volume is reached (V_{min}), the tank is filled with brackish water from the well up to the maximum volume (V_{max}) previously chosen. The process then repeats until finishing the desalination operation.

In the mixing tank salinity controlling operation, a minimum and a maximum allowed mixing tank salinity is set as the operation limits. The salinity of the water in the mixing tank ($c_{s,mix}(t)$) is continuously increasing with the distillate production, while its volume ($V_{mix}(t)$) decreases since there is no brackish water flow from the well. At the instant the maximum salinity allowed is reached ($C_{s,max}$), the pump from the well is turned on to feed the mixing tank until the salinity of the water contained in it falls to the minimum predefined value.

5 Results

5.1 Algorithm Verification

Before analyzing typical well operation scenarios, the developed computational code is verified by comparing its results with available literature data. Since no transient results are available, these comparisons are made with data for the steady-state regime.

Table 2 presents the physical and geometric characteristics of the membranes and modules employed in the comparisons. The experimental results of Wang et al. [2008], Yang et al. [2011] and numerical results of Guan et al. [2014] and Kim [2014] are employed for situations with no heat exchangers.

Table 2: Characteristics (input data) of the membranes and modules used for comparison.

Parameter	**Symbol**	**Unit**	**Wang *et al.* (2008)**	**Kim (2014)**	**Yang *et al.* (2011)**	**Guan**
			Experimental	Numerical	Experimental	Numc
Membrane inner radius	r_l	mm	0.3	0.3	0.46	0.49
Membrane thickness	t	mm	0.11	0.11	0.275	0.240
Polymer thermal conductivity	k_p	W/(m·K)	0.18[a]	0.18	0.18[a]	0.18[a]
Average pore diameter	d_p	mm	0.17[b]	0.17	0.145/0.191	0.2[b]
Porosity	ε_m	—	0.738	0.738	0.83	0.85[b]
Shell inlet temperature	$T_{e,s}$	°C	30–79.3	30–79.3	35–70	30–80
Lumen inlet temperature	$T_{e,l}$	°C	17.5	17.5	25	23.5–
Total pore pressure	P	kPa	101.325[a]	101.325[a]	101.325[a]	101.3
Shell side flow rate	$\dot{V}_s$	L/min	1.648[c]	1.521	2.5–3.0	1.14–
Lumen side flow rate	$\dot{V}_l$	L/min	0.407[c]	0.407	0.4	0.046
Number of fibers	N_{fbr}	—	30[a]	30	20	22
Vessel radius	r_v	mm	3.241[a]	3.232	4.75	7.5
Vessel length	L	m	0.200	0.200	0.225	0.200
Salt mass fraction shell	$w_{NaCl,s}$	%	3.5[d]	3.5	3.5	7.0
Lumen salt mass fraction	$w_{NaCl,l}$	%	0.0	0.0	0.0	0.0
Cold/hot fluid arrangement	—	—	HOCI	HOCI	HOCI	HICC
Direction of flow	—	—	Co-current	Countercurrent	Countercurrent	Coun
Membrane area	A_{memb}	cm^2	150	152.7	220	180.9
Packing density	φ	%	48[a]	48	50	20.8[a]

[a] Obtained from, or employed the calculation procedure in, Kim [2014]. [b] Assumed in comparison with other works. [c] Calculated as a function of average velocities. [d] Lisboa et al. [2021] used $w_{NaCl,s} = 2.5\%$ in their simulations.

Table 3 presents a verification of the results obtained with the present work together with the work of Lisboa et al. [2021] and a comparison with the experimental results of Wang et al. [2008].

The results show that the present methodology is consistent with the analyzed physics, at least in

Table 3: Verification of the computational code with the numerical results of Lisboa et al. [2021] and comparison with the experimental data of Wang et al. [2008]. $T_{e,l} = 17.5°C$.

$T_{e,s}$ [°C]	ΔT [°C]	$T_{o,s}$ [°C]		$T_{o,l}$ [°C]		j_{wl} [kg/(m^2·h)]	
		Present	Wang (2008)	Present	Wang (2008)	Present	Wang (2008)
40.0	22.5	38.5	39.0	23.4	24.5	5.5	6.2 ± 0.4
49.5	23.0	47.2	48.0	26.5	27.8	10.1	11.5 ± 0.7
59.8	42.3	56.2	57.4	31.7	31.3	20.0	19.9 ± 1.0
70.2	52.7	65.1	67.2	37.5	36.2	32.8	30.8 ± 1.0
79.3	61.8	72.6	76.0	43.8	39.0	47.9	41.5 ± 1.2

* Based on the outer radius of the hollow fiber. $T_{Wang_et_al.}(2008)$: $T \pm 0.1°C$.

the steady-state regime, when compared to the experimental results of Wang et al. [2008]. This table also shows that the results obtained with the methodology in transient regime converge, as would be expected for long times, to the numerical results of Lisboa et al. [2021].

Another verification is illustrated in Figure 4, which shows a comparison of the results obtained in the present work for the produced distillate flux with the experimental results of Wang et al. [2008] and the numerical results of Kim [2014].

Figure 4 – Comparison between the result obtained with the present methodology and the experimental results of Wang e

Figure 4: Comparison between the result obtained with the present methodology and the experimental results of Wang et al. [2008] and numerical results of Kim [2014] for the distillate flux.

To illustrate the spatial temperature distribution along the DCMD module, Figure 5 brings a comparison between the results obtained with the present formulation and the numerical results of Guan et al. [2014] for the steady-state temperatures along the feed channel, T_s, of the feed/membrane interface, $T_{m,s}$, of the membrane/distillate interface, $T_{m,l}$, and of the distillate channel, T_l. The inlet temperatures and flow rates of the feed and distillate streams employed in the simulation were: $T_{e,s} = 80°C$, $T_{e,l} = 30°C$ and $\dot{V}_s = 1.1425$ L/min, $\dot{V}_l = 0.04635$ L/min, respectively.

Figure 5 – Temperature distribution along the axial position in the DCMD module. Comparison with Guan et al. (2014)

Figure 5: Temperature distribution along the axial position in the DCMD module in the feed, interface, and distillate regions. Comparison with the numerical results of Guan et al. [2014].

5.2 Simulations with the Brackish Water Well

5.2.1 Preliminary Analysis

For simulation of a real desalination system fed by brackish water wells, data from wells commonly found in the northeastern region of Brazil were used, whose temperatures, salt concentrations, typical flow rates, and other properties can be found in Veiga [2017]. Table 4 provides the characteristic parameter values of 444 wells located in the Ceará state of Brazil.

The parameters relating to the characteristics of the DCMD module and the membrane are those provided in Table 5. The flow arrangement of the module is carried out in the HOCI configuration

Table 4: Characteristics of wells in the state of Ceará/Brazil.

Limits	**Salt concentration [g_s/L]**	**Flow rates [m^3/h]**
Maximum	13	15
Minimum	0.199	0.10
Average	2.532	3.45
Adopted	**3.5**	**3**

$T_{well} = T_{amb} = 25°C$

(hot fluid in the shell and cold fluid in the lumen) in counter-current.

Table 5: Input data, membrane and module, used in the simulation of the wells.

Parameter	**Symbol**	**Unit**	**Lisboa *et al.* (2021)**
Membrane inner radius	r_l	mm	0.3
Membrane thickness	t	mm	0.1
Polymer thermal conductivity	k_p	W/(m·K)	0.2
Average pore diameter	d_p	mm	0.2
Membrane porosity	ε_m	—	0.8
Inlet temperature on shell side	$T_{e,s}$	°C	80
Inlet temperature on the lumen side	$T_{e,l}$	°C	25
Total pore pressure	P	kPa	101.325
Shell-side flow rate	$\dot{V}_s$	L/min	30–50–80
Flow rate on the lumen side	$\dot{V}_l$	L/min	30–50–80
Number of fibers	N_{fbr}	—	10,000
Vessel radius	r_v	mm	50
Vessel length	L	m	1
Salt mass fraction on shell side	$w_{NaCl,s}$	%	0.035
Lumen-side salt mass fraction	$w_{NaCl,l}$	%	0
Membrane area	A_{memb}	m^2	7.2
Packing	φ	%	0.64

To illustrate a typical dynamic behavior of the heat transfer process inside the DCMD module, Figure 6 shows the bulk temperature profiles of the retentate and distillate streams along the respective channels inside the module, at four instants of time (approximately 2.8×10^{-4} s, 3.4 s, 16 s, and 30 s), for the flow condition $\dot{V}_l = \dot{V}_s = 50$ L/min.

Figure 6 – Time evolution of temperature profiles inside the DCMD module.

Figure 6: Time evolution of temperature profiles inside the DCMD module for $\dot{V}_l = \dot{V}_s = 50$ L/min.

As can be seen from this figure, the time evolution of the heat transfer inside the DCMD module is very fast, since the steady-state condition is reached in just a few seconds, taking no more than 30 s for the module to attain this regime.

Figures 7a and 7b illustrate bulk temperature profiles of the feed and permeate channels along the module for different conditions of volumetric flow rate of the streams, considering steady-state single-

pass flow in the DCMD module, and no heat recovery ($\lambda_0 = \lambda = 1$, $\dot{V}_{well} = \dot{V}_s$, $\varepsilon_{HX1} = \varepsilon_{HX2} = 0$).

Figure 7 – Steady-state temperature profiles along the axial position in the module DCMD.

Figure 7: Steady-state temperature profiles along the axial position in the module DCMD for: a) different feed and distillate flow rates; b) equal distillate and feed flow rates ($\dot{V}_l = \dot{V}_s$).

For an intuitive understanding of how the operational and configurational decisions affect the energy efficiency, Christie et al. [2020], based on Lin et al. [2014], introduced a critical mass flow rate ratio, α^*, to define the theoretical condition of maximum water recovery factor in steady-state single-pass DCMD systems, R_{max}:

$$\alpha^* = \left.\frac{\dot{m}_s\, c_{p,s}}{\dot{m}_l\, c_{p,l}}\right|_{\text{critical}}, \quad \alpha = \frac{\dot{m}_s\, c_{p,s}}{\dot{m}_l\, c_{p,l}} \tag{59}$$

$$R_{max}^{FLR} > R_{max}^{DLR} \tag{60}$$

For $\alpha < \alpha^*$, the module operates in the DLR regime. Conversely, for $\alpha > \alpha^*$, the module operates in the FLR regime.

Table 6 brings some results obtained for the averaged theoretical maximum water recovery factor, distillate mass flux, temperature polarization coefficient, and DCMD module thermal efficiency (GOR and SEC_{th}). Three scenarios related to the use of the external heat exchangers for two feed salinity levels and different feed and distillate flow rates were studied.

Table 6: Influence of the operational conditions on the averaged DCMD module efficiency metrics. Steady-state regime (8 h).

Flow Condition	α	α^*	**Oper.**	T_{thr} [°C]	Z_{thr} [m]	RR	$\dot{V}_{jwl}$	TPC	η_{th}	GOR / SEC_{th} [kWh/L]		
							[L/min]	[-]	[%]	$\varepsilon_{HX1,2} = 0$	$\varepsilon_{HX1} = 0.8$; $\varepsilon_{HX2} = 0$	ε_H
$w_{NaCl,s} = 3.5\%$												
$\dot{V}_l = 30 < \dot{V}_s = 80$	0.38	0.88	DLR	80.0	0.63	2.5	2.6	0.73	73	0.28 / 2.5	0.41 / 1.7	
$\dot{V}_l \approx 44 < \dot{V}_s = 50$	**0.88**	**0.88**	**FLR**	—	—	**5.0**	**3.3**	**0.56**	**67**	**0.56 / 1.2**	**1.9 / 0.36**	
$\dot{V}_l = \dot{V}_s = 50$	1.0	0.88	FLR	—	—	5.2	3.4	0.59	66	0.58 / 1.2	1.8 / 0.38	
$w_{NaCl,s} = 12.0\%$												
$\dot{V}_l \approx 42 < \dot{V}_s = 50$	**0.74**	**0.74**	**FLR**	—	—	**2.2**	**1.7**	**0.72**	**37**	**0.30 / 2.3**	**0.88 / 0.79**	
$\dot{V}_l = \dot{V}_s = 80$	0.88	0.74	FLR	—	—	2.9	3.6	0.68	50	0.40 / 1.7	0.96 / 0.72	

(a) Feed-side threshold temperature, $T_{thres,s}$. (b) Distillate-side threshold temperature, $T_{thres,l}$.

5.2.2 Operating Regime Analysis

After characterizing the DCMD desalination system, investigation of the behavior of the system for different operating modes is now performed. In all situations analyzed, the FLR configuration to which $\dot{V}_l = \dot{V}_s = 50$ L/min is adopted.

Figures 8 and 9 illustrate the dynamic behavior of the GOR and SEC_{th} metrics for the batch operation mode and for two different maximum allowed salt concentration of the brine in the mixing tank analyzed, $C_{s,max}$ (120 g_s/L_{mix} and 260 g_s/L_{mix}, respectively). The averaged values indicated in these figures were obtained for an operating time of 8 hours.

Figure 10 illustrates the temporal evolution of the main DCMD system parameters for the different desalination system operating modes (batch, semi-batch, volume-controlling, continuous and steady-

Figure 8 – Evolution of the instantaneous GOR and SECth for $C_{s,max} = 120\ g_s/L_{mix}$.

Figure 8: Evolution of the instantaneous a) GOR and b) SEC_{th}, for $C_{s,max} = 120$ g_s/L_{mix}. Batch operating mode.

Figure 9 – Evolution of the instantaneous GOR and SECth for $C_{s,max} = 260\ g_s/L_{mix}$.

Figure 9: Evolution of the instantaneous a) GOR and b) SEC_{th}, for $C_{s,max} = 260$ g_s/L_{mix}. Batch operating mode.

state): mixture salt concentration in the mixing tank $C_{s,mix}$, volumetric flow rate of the produced distillate $\dot{V}_{jwl}$, mixture volume in the mixing tank V_{mix}, mixture temperature in the mixing tank T_{mix}, volumetric flow rate of the brackish water well $\dot{V}_{well}$, and rejection fraction λ.

Figure 10 – Dynamic evolution of the instantaneous desalination system parameters for different operating modes.

Figure 10: Dynamic evolution of the instantaneous desalination system parameters for different operating modes and $C_{s,max} = 120$ g_s/L_{mix}: a) $C_{s,mix}$, b) $\dot{V}_{jwl}$, c) V_{mix}, d) T_{mix}, e) λ, and f) $\dot{V}_{well}$.

According to these figures, the cycling time for the batch mode is about 1789 s ($\approx$ 30 min) while for the V-control and semi-batch modes are 4800 s ($\approx$ 1 h 20 min) and 6090 s ($\approx$ 1 h 41 min), respectively.

In an attempt to identify the most appropriate operating mode of the DCMD desalination system, comparisons are performed in Figures 11 and 12, where the averaged specific thermal energy cost, SEC_{th}, and water recovery ratio, RR, are displayed, respectively.

Overall, according to these figures, one must seek a balance between the specific energy cost and the water recovery ratio, since there exist, for the same maximum operational salinity, a trade-off between these metrics. When it comes to the energy cost, and for the same limiting mixture salinity, the best operating mode is the batch, followed by the semi-batch, steady-state, and continuous modes, as indicated in the literature [Swaminathan and Lienhard, 2018]. On the other hand, when it comes to water recovery, Figure 12 reveals that the best choice is the semi-batch operating mode, followed by the V-control and batch modes.

Figure 13 shows the potential of each operating mode in producing distillate water when the desalination system is operated for 8 h.

In general, the desalination system can produce up to 1,200 L of distillate water running in any of the operating modes for the lower salinity level. On the other hand, when the salinity increases, it impacts negatively, especially for the steady-states modes (continuous and steady-state), when only about 750 L of distillate water is produced.

Therefore, if the desalination system must be implemented in areas with hydrologic seasonal stress, as the semi-arid region of the Brazilian northeastern, the results point out that the semi-batch could be the preferred choice for brackish water desalination, since the main interest is in saving salt water from the well.

6 Conclusions

The present work developed a mathematical modeling and computer simulation of a DCMD desalination system with double heat recovery from the distillate stream and the feed stream, together with

Figure 11 – Effect of the system operation mode on the averaged specific thermal energy cost.

Figure 11: Effect of the system operation mode on the averaged specific thermal energy cost, SEC_{th}, for different mixture salinity.

Figure 12 – Effect of the system operation mode on the water recovery ratio.

Figure 12: Effect of the system operation mode on the water recovery ratio, RR, for different mixture salinity.

total or partial recirculation of the retentate. The main goal was to employ the developed code to identify the best scenarios for operation of the system in terms of maximizing its thermal efficiency at the same time minimizing the use of brackish water from wells, seeking optimized production of the amount of distilled water.

In general, the following conclusions can be summarized from the present results:

- Heat recovery from the distillate stream already produces a significant increase in the energy efficiency of the desalination system, being a mandatory condition for operation of DCMD-based systems.
- Use of an additional heat exchanger to recover heat from the retentate stream also improves the energy efficiency of the process, especially when the system operates in the FLR condition with mass flow rate near the critical mass flow rate.
- Peaks in GOR values and lows in SEC values can be observed when the well and an external heat exchanger coupled to its stream operate simultaneously, so double heat recovery and partial recirculation of the retentate play an important role in the competitiveness of the membrane distillation process compared to other forms of desalination, especially in long operation processes.
- The amount of heat supplied to the system is strongly impacted by the rejection fraction of the retentate stream.
- The system operating mode has a strong impact on the energy consumption and water recovery of the DCMD desalination system.
- The lowest specific thermal energy cost is associated with the batch operation mode of the desalination plant, however, in regions with hydrologic seasonal stress, the results point out that the semi-batch could be the preferred choice for brackish water saving. Therefore, its choice must be based on information as where the system has to be implanted and what are the availability of energy and brackish water salinity.

Figure 13 – Effect of the system operation mode on the produced volume of distillate water.

Figure 13: Effect of the system operation mode on the produced volume of distillate water, for different mixture salinity.

Extension to other desalination MD methods, like AGMD and V-AGMD, are currently in development, in order to simulate the entire experimental TRL-6 desalination unit installed in the Lab-MEMS/UFRJ.

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